Alec de Moraes

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Education

University of Michigan — **Expected: May 2027**

Bachelors of Science (BS), Computer Engineering — *Ann Arbor, MI*

- Coursework: Electronic Circuits, Signals and Systems, Logic Design, Data Structures and Algorithms, Computer Organization, Digital Integrated Circuits, Embedded Control Systems, VLSI Design I

Work Experience

Hardware Engineer Intern — **May 2025 – Aug 2025**

Dedicated Freight Handlers — *Murrieta, CA*

- Developed bare-metal firmware in C for resource-constrained microcontrollers, implementing 2 low-level drivers (UART, I2C) that enabled end-to-end data collection and transmission of business data on prototype hardware.
- Created Python test scripts for hardware validation of embedded systems, automating test workflows and reducing manual validation time while enabling scalable RTOS-based firmware solutions.

Activities

M-STARX — **Jan 2025 – Present**

University of Michigan — *Ann Arbor, MI*

- Designed custom wiring, a wearable LED sign subsystem, and tested/wired electrical components for an exoskeleton aimed at assisting first responders and competing against other exoskeletons in competition.
- Co-authored and presented a Critical Design Review to professors and sponsors, communicating the exoskeletons architecture and integration plan to earn actionable guidance.

Projects

Stock Market Simulation | *C++* — **Sep 2025 – Oct 2025**

- Programmed an event-driven stock market simulation in **C++** using STL priority-queues to track stock pricing statistics and trader activity as well as analyze historically optimal buy/sell prices.

Traffic Light Controller | *Verilog* — **Mar 2025**

- Created a synchronous finite state machine traffic light controller for a 3-way intersection using sensor inputs to coordinate traffic flow
- Implemented 7-segment display logic to show the remaining time and light color for each lane and verified behavior via test-bench and simulation.

Sequential Calculator | *Verilog* — **Mar 2025 – Apr 2025**

- Designed and implemented a four-function calculator in **Verilog** using RTL design principles, with a custom datapath/controller, overflow detection, and signed-magnitude/two's complement handling.
- Verified functionality through ModelSim simulation and hardware testing on an FPGA board.

Hand Movement Replication | *Microcontrollers, HTML, JavaScript* — **Feb 2026**

- Programmed an ESP32 microcontroller to read in accelerometer data via SPI/I2C and compute real-time finger bend and wrist rotation movements, mapping sensor orientation to motion outputs.
- Drove a synchronized robotic hand with servo motors for finger tendon movement and a stepper motor for wrist rotation. Added an ESP32 hosted web-server coded with HTML and JavaScript for visual indication of finger and wrist movement.

Skills

Languages: C/C++, Python, Verilog, ARM, MATLAB
General: Embedded C/C++, Embedded Systems, Microcontrollers, UART, I2C, SPI, RTOS, Firmware Development, Device Drivers, Cadence Design Systems, PCB Design, ModelSim, Object-oriented programming, Git, Soldering, Electrical Datasheets, Oscilloscopes, Multimeters